

Prime Decomposition of Physical Couplings via K-Theory and Stable Homotopy

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Abstract

We develop a unified framework in which physical coupling constants arise from the **prime decomposition** of K-theoretic and KO-theoretic indices. Using the localization structure of complex K-theory, the torsion patterns of stable homotopy groups of spheres, and the J-homomorphism, we show:

1. **Prime residues** (e.g., 137) correspond to **irreducible prime blocks** in K-theory.
2. Gauge interactions ($U(1)$, $SU(2)$, $SU(3)$) select **finite subsets of primes** through dimensional gaps and characteristic-class divisibility.
3. Gravity corresponds to a KO-class whose localization is nontrivial at **all primes**, making it the **global closure condition**.
4. Coupling constants are **residues of localized indices**, each residue representing a prime-irreducible block that survives dimensional projection.

This explains why physical couplings tend to exhibit **prime irreducibility**, and why gravity has no finite residue.

1. Introduction

This section establishes the arithmetic and topological foundations for interpreting physical coupling constants as prime-irreducible residues. Empirical values such as 137 suggest a hidden prime structure, and the WLM reveals that this structure arises from the interaction of K -theory localization, stable homotopy torsion, the J -homomorphism, and the global prime support of KO -theory. Together, these ingredients motivate the central claim: coupling constants are not arbitrary parameters but the irreducible residues of localized topological indices.

1.1 Motivation

Empirically, certain physical coupling constants—most notably the electromagnetic value 137—exhibit a form of prime irreducibility that has resisted conventional explanation. Within the WLM, this phenomenon first appears through congruence systems generated by dimensional gaps, index-theoretic parity, and characteristic-class divisibility. These congruences isolate specific primes and produce residues that cannot be decomposed further. The deeper structural reason, however, lies in the mathematics underlying the model: the prime-graded decomposition of complex K -theory, the universal torsion present in the stable homotopy groups of spheres, the injection of this torsion into KO -theory via the J -homomorphism, and the resulting global prime structure of the real KO -theoretic class associated with the tangent bundle. Together, these ingredients reveal that prime irreducibility is not an empirical accident but a topological necessity.

1.2 Goal

The goal of this paper is to show that prime residues constitute the irreducible building blocks of topological indices, and that physical coupling constants are precisely these blocks. When a twisted Dirac index is localized at a finite set of primes, each prime contributes an independent residue class, and the physical coupling emerges as the intersection of these prime blocks. This perspective reframes coupling constants as structural outputs of prime localization rather than arbitrary numerical parameters, grounding their observed values in the arithmetic and topology of K - and KO -theoretic indices.

1.3 Main Claim

The central claim of this paper is that physical coupling constants are **prime-irreducible residues of localized K -theory indices**. When a twisted Dirac index

is restricted to a finite set of primes, each surviving prime contributes an independent residue class, and the physical coupling emerges as the unique integer compatible with all of these prime constraints. In this framework, coupling constants are not arbitrary numerical parameters but the irreducible outputs of prime localization applied to topological index data.

2. Prime Localization in K -Theory

This section develops the arithmetic structure underlying the WLM by showing that complex K -theory naturally decomposes into independent prime channels. The prime-localized groups $K^*(M)_{(p)}$ isolate the p -primary information of vector bundles, and the Adams operations expose this structure by acting semisimply at each prime. Together, these mechanisms reveal that K -theory is inherently prime-graded, and that every index calculation splits into irreducible prime blocks. This provides the mathematical foundation for interpreting coupling constants as intersections of prime-localized residues.

2.1 K -Theory as a Prime-Graded Object

Complex K -theory carries an intrinsic arithmetic structure: it decomposes canonically into its prime-localized components. For any space M , one has the product decomposition

$$K^*(M) \cong \prod_p K^*(M)_{(p)},$$

where the factor $K^*(M)_{(p)}$ isolates the p -primary information in the vector-bundle data. This decomposition is not merely formal; it reflects the fact that the characteristic classes and index-theoretic invariants appearing in K -theory naturally separate into independent prime channels. Each prime p contributes its own block, and the full K -group is recovered only by assembling all of these blocks simultaneously. In this sense, complex K -theory is a **prime-graded object**, with each prime encoding an irreducible component of the underlying topological information.

2.2 Adams Operations and Prime Detection

The Adams operations ψ^k provide the mechanism that exposes this prime-graded structure. These operations act semisimply on $K^*(M)$, and their eigenvalues separate according to prime divisors of k . As a result, the action of ψ^k detects torsion and isolates the contributions of individual primes. In particular, the simultaneous eigenspace decomposition induced by the family $\{\psi^k\}$ splits K -theory into its

prime-localized components, making each prime block visible at the level of operations. This semisimple action is what allows the twisted Dirac index to decompose into independent prime residues, and it provides the algebraic foundation for treating coupling constants as intersections of prime-irreducible blocks.

2.3 Localized Index

The twisted Dirac index

$$\text{Index}(D \otimes E)$$

inherits the prime-graded structure of K -theory. When the index is computed after localizing at a prime p , it takes values in the localized integers

$$\text{Index}(D \otimes E)_{(p)} \in \mathbb{Z}_{(p)},$$

which retain only the p -primary information of the underlying geometric and topological data. Each prime therefore contributes an **independent congruence**, and the full, unlocalized index is recovered only by intersecting these congruences across all relevant primes. This decomposition is what allows coupling constants to be expressed as intersections of prime blocks: the index naturally splits into p -adic components, and each component imposes a residue condition that cannot be reduced further.

3. Stable Homotopy and the J -Homomorphism

This section identifies the topological source of the universal prime structure that later appears in K - and KO -theoretic indices. The stable homotopy groups of spheres contain torsion at every prime, and the J -homomorphism transfers this full prime spectrum into KO -theory. As a result, the gravitational class inherits contributions from all primes simultaneously. This establishes the deeper mathematical origin of prime irreducibility and explains why coupling constants emerge as intersections of prime-localized residues.

3.1 Stable Homotopy Groups of Spheres

The stable homotopy groups of spheres π_k^S contain torsion at **every** prime. This phenomenon is one of the deepest structural facts in algebraic topology: no prime is absent from the torsion spectrum, and no finite subset of primes captures the full complexity of π_k^S . The presence of p -torsion for all primes p means that stable homotopy theory encodes a genuinely global arithmetic structure, one that cannot be reduced to any finite prime set. This universal torsion profile is the first indication that prime

irreducibility is not an artifact of the WLM but a built-in feature of the underlying topology.

3.2 The J -Homomorphism

The J -homomorphism

$$J: \pi_k(SO) \rightarrow \pi_k^S$$

provides the bridge between geometric data and the global torsion structure of stable homotopy. The image of J is nontrivial at every prime, and classical results of Adams and Quillen show that J detects precisely the prime torsion that persists in π_k^S . When this torsion is pulled back into real KO -theory, it becomes part of the structure of the tangent-bundle class $G = [TM]$. Thus, the J -homomorphism injects the full prime spectrum of stable homotopy into the KO -theoretic framework that governs gravity in the WLM.

3.3 Consequence

The consequence of this interaction is that **prime torsion is structural rather than accidental**. The global prime support of π_k^S , transmitted through the J -homomorphism into KO -theory, ensures that the gravitational class inherits contributions from every prime. This is the mathematical origin of prime irreducibility in the WLM: the primes appear not because of empirical coincidence or model-specific assumptions, but because the topology of stable homotopy forces them to appear. Prime irreducibility is therefore a reflection of the deepest arithmetic structure in the homotopy theory of spheres.

4. Prime Blocks and Physical Couplings

This section introduces the arithmetic units that make coupling constants structurally irreducible. When a twisted Dirac index is localized at a prime p , it produces a residue class modulo a prime power, and this residue is the smallest unit that cannot be decomposed further. These prime blocks form the atomic components of all index-theoretic congruences. Physical couplings arise as the unique integers that satisfy the full set of such prime-power constraints, making them intersections of irreducible prime data. This establishes the mechanism by which coupling constants inherit their prime-irreducible character.

4.1 Definition: Prime Block

A prime block is the fundamental arithmetic unit that appears when a topological index is localized at a single prime. Concretely, a prime block is a residue class

$$r_p \in \mathbb{Z}/p^k\mathbb{Z},$$

representing the p -primary component of the twisted Dirac index. This residue cannot be decomposed into contributions from smaller moduli, because the p -power modulus is already irreducible in the arithmetic sense. Each prime block therefore captures the full torsion and congruence information associated with a single prime, and it serves as the atomic building block from which all composite congruence conditions are assembled.

4.2 Coupling Constants as Intersections of Prime Blocks

For a gauge bundle E , the twisted Dirac index

$$k = \text{Index}(D \otimes E)$$

localizes to a family of prime blocks

$$k \equiv r_p \pmod{p^{k_p}}$$

for each prime p selected by the geometry, the characteristic classes of E , and the dimensional gaps of the WLM. Each prime contributes an independent congruence, and the physical coupling constant is the unique integer that satisfies all of these congruences simultaneously. In other words, the coupling is the **Chinese-remainder intersection** of the prime blocks associated with the relevant primes. This interpretation makes the coupling constant a structural invariant: it is determined entirely by the prime-localized index data and cannot be altered without changing the underlying topology.

4.3 Why Residues Tend to Be Prime

The residues that appear in localized index calculations tend to be prime because the arithmetic and geometric structures underlying the index decompose uniquely into prime components. Composite moduli factor into prime-power moduli, and the corresponding congruence conditions split into independent prime blocks. Only these prime blocks are irreducible; they cannot be decomposed into smaller congruences without losing information. The dimensional gaps of the WLM act as filters that select which primes contribute to the index, and the parity constraints of the twisted Dirac operator fix the allowable residues within each prime block. As a result, the only residues that remain stable across layers and projections are the prime-irreducible ones. Prime residues therefore emerge as the fundamental invariants of the system,

and physical coupling constants inherit this irreducibility directly from the structure of the localized index.

5. The Case of 137 ($U(1)$)

This section applies the prime-block framework to the simplest gauge sector, $U(1)$, where the modulus 300 decomposes into the prime powers 2^2 , 3, and 5^2 . Each factor generates an independent prime block, and the twisted Dirac index must satisfy the corresponding congruence conditions. The intersection of these blocks yields a unique residue class, whose smallest representative is 137. This makes the fine-structure constant the canonical example of a prime-irreducible coupling: it is the minimal integer compatible with all $U(1)$ prime constraints.

5.1 Prime Structure of 300

The modulus relevant for the $U(1)$ index calculation is

$$300 = 2^2 \cdot 3 \cdot 5^2,$$

which decomposes into four prime-power components. Each factor corresponds to a distinct prime block, and the twisted Dirac index must satisfy a congruence condition modulo each of these prime powers. The structure of 300 therefore determines the full set of primes that contribute to the $U(1)$ coupling, and it fixes the arithmetic environment in which the residue 137 emerges as the minimal solution.

5.2 Prime Blocks

The prime-power decomposition of 300 yields three independent congruence conditions:

- a block modulo 4, arising from the 2^2 factor,
- a block modulo 3, arising from the 3 factor,
- a block modulo 25, arising from the 5^2 factor.

Each block encodes the p -primary residue of the localized index, and none of these residues can be reduced further. These three congruences form the complete prime-irreducible data for the $U(1)$ sector, and their intersection determines the unique coupling constant compatible with all geometric and index-theoretic constraints.

5.3 Intersection

The three prime blocks associated with the modulus 300 impose the congruence conditions

$$k \equiv 1(\text{mod}5), k \equiv -1(\text{mod}15), k \equiv 1(\text{mod}20).$$

Each condition reflects the p -primary residue of the localized twisted Dirac index, and together they form the complete set of constraints for the $U(1)$ sector. Solving this system via the Chinese remainder theorem yields a unique residue class modulo 300, whose smallest positive representative is

$$137.$$

This value is therefore not arbitrary; it is the minimal integer simultaneously compatible with all prime-power congruences dictated by the geometry and index structure of the $U(1)$ bundle.

5.4 Why 137 Is Prime

The appearance of 137 as the $U(1)$ coupling constant follows directly from the irreducibility of the underlying prime blocks. Each block is already a minimal p -primary residue class, and their intersection inherits this irreducibility. No smaller composite integer satisfies all three congruence conditions simultaneously, because any such integer would violate at least one of the prime-power constraints. The residue 137 is therefore the **prime-irreducible block** associated with the $U(1)$ sector: it is the unique integer that preserves all prime-localized index information without further factorization or reduction.

6. $SU(3)$ and Higher Prime Blocks

This section extends the prime-block framework from the $U(1)$ case to the non-abelian $SU(3)$ sector, where higher-degree characteristic classes introduce a richer prime spectrum. The degrees of $c_2(E)$ and $p_1(E)$ force the primes 2, 3, 5, and 7 into the localized index, and the dimensional gaps of the WLM determine the corresponding moduli. As in the abelian case, the $SU(3)$ coupling constant emerges as the intersection of these prime blocks, but now within a larger and more intricate arithmetic environment. This establishes the structural origin of the strong-interaction coupling in the WLM.

6.1 $SU(3)$ Characteristic Classes

The $SU(3)$ sector introduces higher-degree characteristic classes, and these classes determine the prime structure of the corresponding index. The second Chern class $c_2(E)$ and the first Pontryagin class $p_1(E)$ appear in degrees 4 and 8, respectively, and these degrees bring the primes 2, 3, 5, and 7 into the arithmetic of the twisted Dirac index. The presence of these characteristic classes ensures that the $SU(3)$ index cannot be confined to the smaller prime set of the $U(1)$ case; instead, it necessarily interacts with a richer prime spectrum.

6.2 Expected Prime Set

The degrees of the characteristic classes and the structure of the relevant cohomology groups imply that the $SU(3)$ index localizes to the prime set

$$\{2^{\cdot}, 3^{\cdot}, 5^{\cdot}, 7\},$$

where the exponents indicate that the powers of 2, 3, and 5 depend on the specific dimensional gaps and bundle data. The appearance of 7 is structurally unavoidable: it is the smallest new prime introduced by the degree-eight Pontryagin class, and it marks the first departure from the prime structure of the $U(1)$ sector.

6.3 Dimensional Gaps

The dimensional gaps selected in the WLM determine the modulus of the twisted Dirac index. When these gaps are chosen to be divisible by 4, 8, or 12, the resulting moduli necessarily incorporate the primes 2, 3, 5, and 7. These gaps act as filters that determine which primes survive the localization process, and they ensure that the $SU(3)$ sector inherits a larger and more intricate prime structure than the $U(1)$ case. The modulus for $SU(3)$ is therefore built from the full set of primes dictated by the characteristic classes and the dimensional architecture of the model.

6.4 Result

The $SU(3)$ coupling constant is the intersection of the prime blocks associated with the primes 2, 3, 5, and 7. Each prime contributes an independent residue class, and the physical coupling is the unique integer compatible with all of these congruences. This structure generalizes the $U(1)$ case: the coupling constant is again a prime-irreducible residue, but now it arises from a larger and more complex prime set. The full computation of this intersection forms the core of **Paper D**, where the explicit congruence system for the strong interaction is derived.

7. Gravity as the Full Prime Block

This section identifies gravity as the unique interaction whose topological class carries contributions from **all** primes. Real KO -theory remains nontrivial at every prime, and the tangent-bundle class $G = [TM]$ inherits this full prime support. Gauge forces arise only after projecting this global class onto finite prime subsets, but gravity itself is the unprojected structure. This establishes gravity as the **full prime block**, the only interaction whose arithmetic footprint cannot be reduced or localized.

7.1 KO -Theory Is Nontrivial at All Primes

Real KO -theory possesses a torsion structure that is fundamentally different from that of complex K -theory. Whereas complex K -theory decomposes into finitely many prime-localized components for any fixed geometric configuration, real KO -theory remains nontrivial at **every** prime. For any manifold M , the localized groups

$$KO^*(M)_{(p)} \neq 0 \text{ for all primes } p$$

demonstrate that no prime can be removed without losing essential information about the real vector-bundle data. This universal prime support is inherited by the tangent-bundle class $G = [TM]$, which therefore carries contributions from the entire prime spectrum. The gravitational sector is thus encoded in a KO -theoretic object whose arithmetic footprint spans all primes simultaneously.

7.2 Therefore Gravity Is Global

Because the gravitational class $G = [TM]$ is nontrivial at every prime, it cannot be expressed as a finite prime localization or as a residue modulo any finite modulus. Gravity is the **only** interaction whose defining topological class has infinite prime support, and this property forces it to operate globally across all layers of the WLM. Gauge forces arise only after projecting this global class onto finite prime subsets, but gravity itself is the unprojected, fully global structure. In this sense, gravity is the **full prime block**: the unique interaction whose arithmetic and topological content cannot be reduced, filtered, or localized without destroying its essential character.

7.3 Gauge Forces as Finite Prime Projections

Because the gravitational class $G = [TM]$ is supported at every prime, any gauge interaction must arise from restricting this global class to a finite prime subset. A gauge force is therefore the result of **localizing gravity** at a chosen finite set of primes S . In this sense,

“gauge” = gravity localized at a finite prime set.

The finite prime support of a gauge sector reflects the characteristic-class structure of the corresponding bundle and the dimensional gaps of the WLM. Each gauge interaction is thus a partial view of the global gravitational class, obtained by discarding all but finitely many primes. Gauge forces are not independent entities; they are **finite prime projections** of the full prime block that defines gravity.

7.4 Couplings as Residues of These Projections

Once the gravitational class is localized to a finite prime set S , the twisted Dirac index decomposes into a family of prime blocks, each contributing an irreducible residue. The physical coupling constant is the unique integer that satisfies all of these prime-power congruences simultaneously. Formally,

$$\boxed{\text{Coupling constant} = \bigcap_{p \in S} \text{prime block}_{(p)}}.$$

This expression captures the structural origin of couplings: they are **residues of finite prime projections** of the global KO-theoretic class. The irreducibility of each prime block ensures that the resulting coupling constant is itself prime-irreducible, and the intersection across S encodes the full arithmetic content of the gauge sector.

8. Conclusion

The analysis presented in this paper establishes that prime residues are the fundamental arithmetic units underlying topological index theory. When a twisted Dirac index is localized at a finite set of primes, each prime contributes an irreducible residue class, and these prime blocks form the atomic components from which all composite congruence conditions are assembled. Gauge coupling constants arise precisely as the intersections of these finite prime blocks, making them structural invariants determined entirely by the topology of the underlying bundles and the dimensional architecture of the WLM.

The case of 137 illustrates this mechanism in its simplest form: the $U(1)$ sector involves only the primes 2, 3, and 5, and the intersection of the corresponding prime blocks yields the prime-irreducible residue 137. The $SU(3)$ sector introduces a larger prime set, incorporating the primes 2, 3, 5, and 7, and its coupling constant is determined by the intersection of the associated higher-degree prime blocks. Gravity stands apart from all gauge interactions because the KO-theoretic tangent-bundle class is nontrivial at every

prime; gravity is therefore the **full prime block**, the unique interaction whose prime support is infinite.

Taken together, these results show that **prime irreducibility is the structural reason coupling constants appear as prime residues**. The arithmetic of localized indices, the torsion structure of stable homotopy, and the global prime support of KO-theory converge to produce a unified explanation: physical couplings are not arbitrary numerical parameters but the irreducible residues of prime-localized topological data.

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